Cheese!

SMART ATTENDANCE SYSTEM USING FACE RECOGNITION

INTRODUCTION

Traditional attendance methods, manual roll calls, and card-based systems are often inefficient and error-prone. With the rise of artificial intelligence and computer vision, face recognition offers a secure and automated alternative.

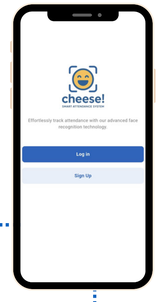
This project introduces a Smart Attendance System that integrates machine-learning-based facial recognition into a mobile application, making attendance tracking easier for educational institutions. The system supports two key roles:

SUPER ADMINS:

- Oversee system-wide operations, manage user accounts, and monitor statistics.

USERS:

- Handle group-level attendance, enrol students, and generate reports.



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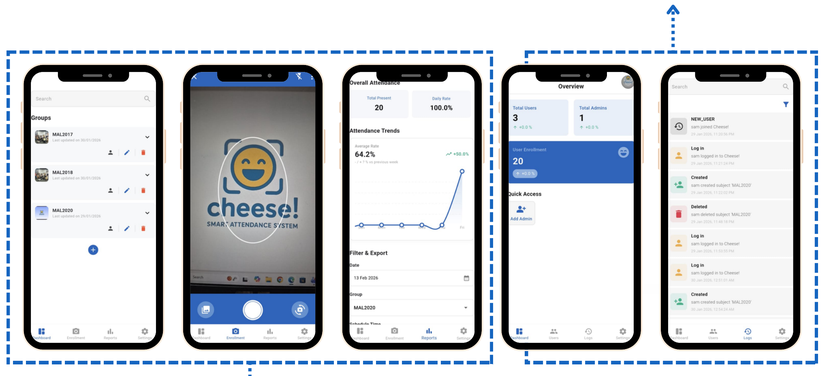
ROLE-BASED ACCESS

SUPER ADMIN

- Full administrative control over the system.
- Can add, edit, or delete user accounts.
- Monitors statistics such as total users, super admins, and face enrollments.
- Receives notifications for all system activities to ensure transparency.



USERS

- Create and manage groups with scheduled sessions (e.g., classes).
- Enrol students' facial data for attendance tracking.
- Conduct attendance during sessions using real-time face recognition.
- Access detailed reports and statistics for each group.
- Customise personal profile settings.

SYSTEM WORKFLOW

Authentication
- Role-based login for super admins and users

Enrollment
- Capture and preprocess student facial images

Attendance
- Real-time recognition and automatic marking

Reporting
- Generate dashboards and statistical insights

CONCLUSION

The **Smart Attendance System** provides a **modern**, **efficient**, and **secure** solution for managing **attendance**. It **reduces human error**, **saves time**, and **enhances transparency** through role-based access and automated reporting.

FUTURE IMPROVEMENT

- Improved attendance-taking performance
- Integration with university databases and LMS
- Multi-factor authentication and access control
- Application in campus security and event management